Cassels–Swinnerton-Dyer conjecture for cubic surfaces

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Abstract

This note resolves the Cassels–Swinnerton-Dyer conjecture in dimension two: a cubic surface over an arbitrary field with a zero-cycle of degree one has a rational point, without any smoothness, reducedness, or geometric integrality assumption. The essential step is a descent from a linear Pfaffian presentation to a rational point in characteristic zero. A degree-two incidence of common isotropic three-spaces, together with a radical vector for a fourth alternating form, produces a point over an extension of degree at most two. Quadratic secant descent and specialization then give a point over the ground field. The classical five-point Pfaffian construction and Voisin's degree reduction prove the smooth case in characteristic zero. Following Ma's lifting method, finite subschemes and mixed-characteristic specialization extend the smooth case to arbitrary fields. A multiplicity argument then handles singular support: a point of degree prime to three either gives a rational point directly or can be replaced by a smooth point over the same extension. Lifting that point to a smoothing and specializing removes the smoothness assumption.

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1 Introduction

The Cassels–Swinnerton-Dyer conjecture predicts that a cubic hypersurface which acquires a rational point over a finite extension of degree prime to three already has a rational point over its ground field. Equivalently, a zero-cycle of degree one should imply a rational point. This asks whether the numerical condition that the index be one is sufficient for the existence of a point. We prove the conjecture for all cubic surfaces, including reducible and nonreduced ones. Of course, it is the smooth case that is essential.

Theorem 1.1. *Let k be an arbitrary field, let $F \in k[x_0, x_1, x_2, x_3]$ be a nonzero homogeneous cubic, and put $X = V(F) \subset \mathbf{P}^3_k$. If X has a zero-cycle of degree one, then $X(k) \neq \varnothing$. Equivalently, if $X(L) \neq \varnothing$ for a finite extension L/k of degree prime to three, then $X(k) \neq \varnothing$.*

Here zero-cycles have integral coefficients and need not be effective. The class

$$h_3 = c_1(\mathcal{O}_X(1))^2 \cap [X] \in \mathrm{CH}_0(X)$$

has degree three, with the fundamental cycle $[X]$ retaining any component multiplicities. A zero-cycle of degree one has in its support a closed point of degree prime to three. Conversely, such a point and h_3 give a zero-cycle of degree one by Bézout's identity. An L-point pushes forward to an effective zero-cycle of degree $[L : k]$, which proves the asserted equivalence. For a smooth surface S we also write $h = c_1(\mathcal{O}_S(1))$ and $h_3 = h^2$.

The principal earlier reduction is due to Daniel Coray. Over a perfect field, a smooth cubic surface with a closed point of degree prime to three has a closed point of degree 1, 4, or 10 [5, Theorem 7.1]. Coray also proved the conjecture over local fields [5]. Qixiao Ma removed the perfectness hypothesis from the degree reduction, proving the same $1, 4, 10$ conclusion for smooth cubic surfaces over arbitrary fields [9, Theorem 2.6]. His result also applies to regular surfaces, which need not be smooth over imperfect fields; this additional generality is not needed for our argument. His method lifts closed points as finite subschemes over a Cohen ring, thereby retaining their degrees even when the residue extensions are inseparable. Jean-Louis Colliot-Thélène developed the relation between such degree reductions and effectivity in the Chow group of zero-cycles on del Pezzo surfaces [4].

Claire Voisin recently sharpened Coray's reduction in characteristic zero: a smooth cubic surface with a zero-cycle of degree one has a closed point of degree 1 or 4 [12, Theorem 1.5]. Her proof uses rank-two vector bundles and effectivity results for zero-cycles. The relation between degrees four and five is already present in her proof of Proposition 4.3(b). In a related direction, Francesca Balestrieri proved that a cubic hypersurface with a point over a simple extension of degree four has a point over an extension of degree one or five [1, Theorem 3.8]. For cubic surfaces in characteristic zero, Proposition 2.3 below gives both directions by intersecting the surface with a split twisted cubic.

Our additional arithmetic input is the following statement.

Theorem 1.2. *Let k have characteristic zero. If a nonzero cubic form $f \in k[x_0, x_1, x_2, x_3]$ has a linear 6×6 Pfaffian presentation over k, then the cubic $f = 0$ has a k-rational point.*

The geometry underlying this statement is classical. The construction of a Pfaffian presentation from five points in general linear position uses the structure theorem of David Buchsbaum and David Eisenbud [3]; it belongs to the theory of determinantal and Pfaffian hypersurfaces developed by Arnaud Beauville [2, Proposition 7.2 and Remark 7.3(b)] and is made explicit for cubic surfaces by Fabio Tanturri [11, Section 2]. The incidence of common isotropic three-spaces for three alternating forms on a six-dimensional space is generically of degree two, as proved by Frédéric Han [7, Proposition 2.4]. We give a short calculation of this incidence over the ground field. The arithmetic step is to choose a radical vector for the fourth form on one of these isotropic spaces. Four covectors then lie in a three-dimensional annihilator, producing a point of the Pfaffian cubic over an extension of degree at most two. Secant descent gives a rational point, and a one-parameter deformation removes the genericity condition.

Consequently, an effective zero-cycle of degree five on a smooth cubic surface in characteristic zero gives a rational point. The twisted-cubic construction and Voisin's theorem establish the smooth case of Theorem 1.1 in characteristic zero. The passage to arbitrary characteristic follows the smooth-surface case of Ma's lifting method: lift a closed point of degree prime to three as a finite flat subscheme in a smooth projective cubic family, apply the characteristic-zero result, and specialize by properness. Section 5 includes the required lifting proof. John Fogarty's relative smoothness theorem for Hilbert schemes of points [6] provides an alternative to the local complete intersection argument used there.

Finally, we remove the smoothness hypothesis by a separate reduction over $k((t))$. A double point can be replaced by a smooth point over the same extension, using the classical projection construction [8, Section 5]. A triple point over an extension of degree prime to three gives a k-point: Euler's identity suffices outside characteristic three, while in characteristic three we descend a translation direction by trace. Thus only a smooth point needs to be lifted to a smoothing. The smooth theorem over $k((t))$ and properness then give a point on the original surface. This argument makes no integrality or reducedness assumption.

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Remark. I, Valery Alexeev, learned about this problem from a talk given by Claire Voisin at a conference in Princeton in June of 2026. As she explained, the dimension 2 case of the Cassels-Swinnerton-Dyer conjecture is equivalent to the existence of a rational section of a universal cubic surface as in Remark 4.4. I asked GPT-5.5 (then the current version) to find a counterexample to CSD, proposing this universal cubic as a candidate. It made no progress. Once GPT-6 Astra came out, I asked it the same question, and it quickly came up with a section. The initial solution was somewhat convoluted, but after some back and forth the key new idea, for a degree $d = 5$ point, was identified. I asked the GPT to write a paper making this new idea the central point of it. Additional requests were to extend the initial argument from characteristic 0 to an arbitrary field, and then from a smooth cubic surface to an arbitrary cubic form. I specifically instructed the GPT to make sure that all human contributors whose papers were used in the argument are properly credited. Once the paper was ready, I asked Codex to formalize it in Lean modulo [5, 12], which it did in a few hours.

2 Small-degree points and twisted cubics

Throughout Sections 2–4, the field k has characteristic zero. The degree of a finite k-scheme is its length over k. We use the associated effective zero-cycle when taking residual intersections.

Lemma 2.1 (Quadratic secant descent)**.** *Let $X \subset \mathbf{P}^3_k$ be the zero locus of a nonzero cubic form. If X has an effective zero-cycle of degree one or two, then $X(k) \neq \varnothing$.*

Proof. It suffices to consider a closed point P of degree two. Its two conjugates span a k-line ℓ. If $\ell \subset X$, it supplies a rational point. Otherwise $[\ell \cap X] - [P]$ is an effective zero-cycle of degree one. □

Lemma 2.2. *Let P be a closed point of degree $d \in \{4, 5\}$ on a smooth cubic surface S/k.*

1. *If P is contained in a k-plane, then $S(k) \neq \varnothing$.*
2. *If P is not contained in a k-plane, its geometric points are in general linear position: any four of them span $\mathbf{P}^3$.*

Proof. For (1), let Π contain P. Since $h^0(\Pi, \mathcal{O}_\Pi(2)) = 6 > d$, there is a conic $Q \subset \Pi$ containing P. Write $C = S \cap \Pi$. If C and Q have no common component, then $[C \cap Q] - [P]$ is effective of degree $6 - d \leq 2$, and Lemma 2.1 applies. Otherwise the greatest common divisor of their equations over k has degree one or two. In the first case it defines a k-line on S; in the second, the residual factor of the cubic defines such a line.

For (2), the assertion is immediate when $d = 4$. Suppose $d = 5$ and choose a finite Galois splitting field L/k. Normalize representatives $v_1, \dots, v_5 \in L^4$ by a common affine chart defined over k. Their span is L^4, so their space of relations is one-dimensional. In a nonzero relation $\sum c_i v_i = 0$, the set of indices with $c_i = 0$ is Galois invariant. Transitivity forces this set to be empty, proving the assertion. The geometric span descends to k, so a proper span would indeed be contained in a k-plane. $\square$

Proposition 2.3. *A smooth cubic surface S/k has an effective zero-cycle of degree four if and only if it has an effective zero-cycle of degree five. More precisely, a closed point of either degree yields either a rational point or a closed point of the other degree.*

Proof. If $S(k) \neq \varnothing$, multiples of a rational point give both cycles. Otherwise Lemma 2.1 excludes points of degrees one and two; hence an effective zero-cycle of degree $d \in \{4, 5\}$ must be a single closed point P. By Lemma 2.2, its geometric points are in general linear position.

We place P on a *split* twisted cubic over k. Write $E = \kappa(P) = k(\theta)$ and let

$$\nu_3 : \mathbf{P}^1_k \longrightarrow \mathbf{P}^3_k, \qquad [s : t] \longmapsto [s^3 : s^2t : st^2 : t^3].$$

For $d = 4$, choose homogeneous coordinates $P = [a_0 : a_1 : a_2 : a_3]$ with $a_i \in E$. Noncoplanarity means that the a_i form a k-basis of E. Comparing this basis with $1, \theta, \theta^2, \theta^3$ gives a matrix in $\mathrm{GL}_4(k)$ taking $[1 : \theta : \theta^2 : \theta^3]$ to P. Its image of $\nu_3(\mathbf{P}^1)$ is the required curve.

For $d = 5$, let $\theta_1, \dots, \theta_5$ be the conjugates of θ in a Galois splitting field. The five points $[1 : \theta_i : \theta_i^2 : \theta_i^3]$ form a projective frame, by the Vandermonde determinant. The corresponding conjugates of P also form a projective frame. There is a unique projectivity matching these two labelled frames. Both labellings have the same Galois action, so uniqueness makes the projectivity Galois invariant; it therefore descends to k. Again its image of $\nu_3(\mathbf{P}^1)$ is a split twisted cubic Γ containing P.

If $\Gamma \subset S$, the curve supplies a k-point. Otherwise Bézout's theorem gives

$$\deg(S \cap \Gamma) = 9, \qquad [S \cap \Gamma] - [P] \text{ effective of degree } 9 - d. \tag{2.1}$$

Under the standing assumption $S(k) = \varnothing$, this effective cycle is a single closed point, as above. $\square$

Remark 2.4. The effectivity argument in Voisin's proof of Proposition 4.3(b) [12] gives another proof. For $H = \mathcal{O}_S(2)$, one has $h^0(S, H) = 10$ and $c_1(H)^2 = 4h_3$. Add to a degree-d point a disjoint reduced line section. Voisin's Lemma 3.6, applied to the resulting subscheme of length $d + 3 \le 8$, makes

$$4h_3 - ([P] + h_3) = 3h_3 - [P]$$

rationally equivalent to an effective zero-cycle of degree $9 - d$. Her proof explicitly uses $d = 5$; the same argument works for $d = 4$. Proposition 2.3 proves the existence equivalence by the direct intersection (2.1). Moreover, when $\Gamma \not\subset S$, the residual cycle represents the same Chow class. Indeed, let $i : S \hookrightarrow \mathbf{P}^3_k$ be the inclusion and let $\ell \subset \mathbf{P}^3_k$ be a line not contained in S. Since $[\Gamma] = 3[\ell]$ in $\mathrm{CH}_1(\mathbf{P}^3_k)$, refined Gysin pullback gives

$$[S \cap \Gamma] = i^![\Gamma] = 3i^![\ell] = 3h_3 \quad \text{in } \mathrm{CH}_0(S).$$

Consequently,

$$[S \cap \Gamma] - [P] = 3h_3 - [P] \quad \text{in } \mathrm{CH}_0(S).$$

3 Pfaffian cubics have rational points

Let $V = k^6$ and put $\mathcal{T} = (\bigwedge^2 V^*)^3 \simeq \mathbf{A}_k^{45}$. For an alternating form $A \in \bigwedge^2 V^*$, a subspace $W \subset V$ is isotropic if $A|_{W\times W} = 0$.

Lemma 3.1 (The degree-two isotropic incidence)**.** *There is a nonempty open subset $\mathcal{T}^\circ \subset \mathcal{T}$, defined over k, such that the scheme of common isotropic three-spaces $W \subset V$ for a triple (A_0, A_1, A_2) in $\mathcal{T}^\circ(k)$ is finite* étale *of degree two over k.*

This is the ordered-triple form of Han's incidence calculation [7, Proposition 2.4] (who credits Sasha Kuznetsov). The following proof also gives a fixed k-rational triple in $\mathcal{T}^\circ$.

Proof. Consider

$$\mathcal{I} = \{(W, A_0, A_1, A_2) : A_i|_{W\times W} = 0 \ (0 \le i \le 2)\} \subset \mathrm{Gr}(3, V) \times \mathcal{T}.$$

The restriction map $\bigwedge^2 V^* \to \bigwedge^2 W^*$ is surjective. Thus $\mathcal{I}$ is the total space of a rank-36 vector bundle over $\mathrm{Gr}(3, 6)$, and is smooth of dimension 45. The projection $\pi : \mathcal{I} \to \mathcal{T}$ is proper.

Write $V = V_1 \oplus V_2 \oplus V_3$, with $\dim V_i = 2$, and set

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Consider the triple

$$B_0 = \mathrm{diag}(J, J, J), \qquad B_1 = \mathrm{diag}(J, 2J, 3J), \qquad B_2 = \begin{pmatrix} 0 & I_2 & I_2 \\ -I_2 & 0 & D \\ -I_2 & -D & 0 \end{pmatrix}. \tag{3.1}$$

A B_0-isotropic three-space W is Lagrangian. Put $T = B_0^{-1}B_1 = \mathrm{diag}(I_2, 2I_2, 3I_2)$. If W is also B_1-isotropic, then

$$B_0(Tw, w') = B_1(w, w') = 0 \quad (w, w' \in W), \qquad T(W) \subset W^{\perp_{B_0}} = W.$$

Thus W is T-invariant and splits into its intersections with the three eigenspaces. Isotropy bounds the dimension of each summand by one, so $W = \ell_1 \oplus \ell_2 \oplus \ell_3$ with $\ell_i \subset V_i$ a line. This description holds scheme-theoretically: the spectral projectors split invariant subbundles after every base change. The common isotropic locus for B_0, B_1 is therefore $(\mathbf{P}^1)^3$.

Writing $\ell_i = [a_i : b_i]$, the remaining equations are

$$a_1a_2 + b_1b_2 = 0, \qquad a_1a_3 + b_1b_3 = 0, \qquad a_2a_3 + 2b_2b_3 = 0.$$

No a_i vanishes at a geometric solution: if $a_1 = 0$ then $b_2 = b_3 = 0$, contradicting the third equation; the other cases are similar. Set $u_i = b_i/a_i$. The fiber is

$$u_2 = u_3 = -u_1^{-1}, \qquad u_1^2 + 2 = 0, \tag{3.2}$$

and hence is finite and geometrically reduced of degree two.

At these two geometric points the differential of π is an isomorphism, since source and target are smooth of the same dimension and the fiber tangent spaces vanish. Thus π is *étale* there. Properness allows us to remove the image of its non-*étale* locus and shrink around (B_0, B_1, B_2); the resulting morphism is proper and quasi-finite, hence finite *étale*, of degree two. $\square$

Proof of Theorem 1.2. Write

$$f(x) = \mathrm{Pf}\left(\sum_{i=0}^{3} x_i A_i\right),$$

where the A_i are alternating matrices over k. First assume $(A_0, A_1, A_2) \in \mathcal{T}^\circ(k)$. Lemma 3.1 gives an extension L/k of degree at most two and a three-space $W \subset L^6$ isotropic for A_0, A_1, A_2.

The alternating form $A_3|_W$ has a nonzero radical vector $v \in W$, since $\dim W = 3$. Consequently all four covectors

$$A_0 v,\ A_1 v,\ A_2 v,\ A_3 v$$

belong to the three-dimensional subspace $\mathrm{Ann}(W) \subset (L^6)^*$. Choose a nontrivial relation $\sum q_i A_i v = 0$ over L. The alternating matrix $\sum q_i A_i$ is singular, so

$$f(q_0, q_1, q_2, q_3)^2 = \det\left(\sum_{i=0}^{3} q_i A_i\right) = 0.$$

This gives an L-point of $f = 0$. Its image has degree at most two, and Lemma 2.1 gives a k-point.

For an arbitrary triple, use the fixed matrices (3.1) and put

$$A_i(t) = (1-t)A_i + tB_i \quad (0 \le i \le 2), \qquad A_3(t) = A_3, \qquad f_t(x) = \mathrm{Pf}\left(\sum_{i=0}^{3} x_i A_i(t)\right).$$

The generic triple belongs to $\mathcal{T}^\circ$ because its value at $t = 1$ does; moreover $f_t \neq 0$ over $k(t)$ since $f_0 = f \neq 0$. The preceding argument over $k(t)$ gives a $k(t)$-point of $f_t = 0$. Scale its homogeneous coordinates into the discrete valuation ring $k[t]_{(t)}$, with at least one a unit. Reduction modulo t then gives a k-point of $f = 0$. □

The specialization applies to the final rational point; the auxiliary isotropic three-space need not specialize. Smoothness of the Pfaffian cubic is not used in this theorem.

4 Five points and the characteristic-zero theorem

Lemma 4.1 (Buchsbaum–Eisenbud, Beauville, Tanturri)**.** *Let $Z \subset \mathbf{P}^3_k$ be a reduced subscheme of degree five whose geometric points are in general linear position. Every cubic form vanishing on Z has a linear 6×6 Pfaffian presentation over k.*

Proof. Put $R = k[x_0, x_1, x_2, x_3]$. The ideal of five points in general linear position is arithmetically Gorenstein, with resolution

$$0 \longrightarrow R(-5) \longrightarrow R(-3)^5 \longrightarrow R(-2)^5 \longrightarrow R \longrightarrow R/I_Z \longrightarrow 0. \tag{4.1}$$

For completeness, this standard five-point calculation can be checked after extending to $\overline{k}$ and moving the points to $e_0, e_1, e_2, e_3, [1:1:1:1]$. Their ideal is generated by the five independent differences among the six products $x_i x_j$ with $i < j$. In the quotient all mixed monomials of a fixed degree agree, and the Hilbert function is $1, 4, 5, 5, \ldots$. Reducing by the nonzerodivisor $\ell = x_0 + x_1 + x_2 + x_3$, and writing q for the common mixed product, gives

$$x_i^2 = -3q, \qquad x_i x_j = q\ (i \neq j), \qquad x_i q = 0.$$

On the basis x_0, x_1, x_2 of degree one, multiplication into degree two is represented by

$$\begin{pmatrix} -3 & 1 & 1 \\ 1 & -3 & 1 \\ 1 & 1 & -3 \end{pmatrix}, \qquad \det = -16.$$

Thus the socle is one-dimensional in degree two. The ring is Gorenstein, and its five quadratic generators and self-duality give (4.1). Gorensteinness and the graded Betti numbers descend by faithful flatness.

Apply the Buchsbaum–Eisenbud structure theorem [3] over k. The middle map of (4.1) is represented by a 5×5 alternating matrix T of linear forms, and I_Z is generated by

$$q_i = (-1)^{i+1} \operatorname{Pf}(T_{\hat{i}}) \qquad (1 \le i \le 5).$$

If f is a cubic vanishing on Z, write $f = \sum_i \ell_i q_i$ with linear forms ℓ_i over k. Expanding along the last column gives

$$f = \operatorname{Pf}\begin{pmatrix} T & \ell \\ -\ell^{\mathsf{T}} & 0 \end{pmatrix}, \qquad \ell = (\ell_1, \dots, \ell_5)^{\mathsf{T}}. \tag{4.2}$$

This is the construction in [2, Remark 7.3(b)] and [11, Section 2]. The alternating resolution is obtained over k itself; no rationality of the individual geometric points is required. □

Theorem 4.2. *A smooth cubic surface over a field of characteristic zero with an effective zero-cycle of degree five has a rational point.*

Proof. If the cycle has a point of degree one or two in its support, apply Lemma 2.1. Otherwise it is a single closed point P of degree five. The coplanar case is Lemma 2.2(1). In the remaining case Lemma 2.2(2) gives general linear position, Lemma 4.1 supplies a Pfaffian presentation over k, and Theorem 1.2 gives a rational point. □

Corollary 4.3. *A smooth cubic surface over a field of characteristic zero with a zero-cycle of degree one has a rational point.*

Proof. Voisin's theorem [12, Theorem 1.5] gives a point of degree one or four. In the latter case Proposition 2.3 gives an effective zero-cycle of degree five, and Theorem 4.2 applies. □

Remark 4.4. There is also a universal formulation. Let $U \subset \mathbf{P}^{19}_k$ be the smooth-cubic locus, let $\mathcal{S} \to U$ be the universal cubic surface, and put $Y = \operatorname{Sym}^4_U(\mathcal{S})$. Over $k(Y)$ the universal surface has its tautological degree-four cycle. Proposition 2.3 and Theorem 4.2 give a rational section of

$$\mathcal{S} \times_U Y \longrightarrow Y.$$

Thus the construction works over the field of the unordered quartet; no ordering of its geometric points is needed.

5 Specialization to arbitrary fields

We now allow arbitrary characteristic. The required lifting statement is the smooth-surface instance of Ma's method [9, Section 2.2, Lemmas 2.4–2.5]; we include its proof.

Lemma 5.1. *Let k have characteristic $p > 0$, let $S \subset \mathbf{P}^3_k$ be a smooth cubic surface, and let $P \subset S$ be a closed point of degree d. There exist a complete discrete valuation ring R of characteristic zero with residue field k, a smooth cubic surface $\mathscr{S} \subset \mathbf{P}^3_R$ with special fiber S, and a finite flat subscheme $\mathscr{Z} \subset \mathscr{S}$ of degree d with special fiber P.*

Proof. Choose a Cohen ring R with uniformizer p and residue field exactly k [10, Tag 0328]. Lift the coefficients of a defining cubic of S to obtain $\mathscr{S} \subset \mathbf{P}^3_R$. This hypersurface is flat over R and smooth along its special fiber. Its nonsmooth locus is closed and proper over R. If nonempty, its image would contain the closed point of $\operatorname{Spec} R$, a contradiction. Thus $\mathscr{S}/R$ is smooth.

The closed immersion $P \hookrightarrow S$ is a local complete intersection: the maximal ideal of the regular local ring $\mathcal{O}_{S,P}$ is generated by a regular sequence. By the infinitesimal lifting theorem for finite local complete intersections, the relative Hilbert functor is formally smooth at $[P]$ [10, Tag 06D9]. Since the relative Hilbert scheme $\operatorname{Hilb}^d(\mathscr{S}/R)$ is locally of finite presentation, it is smooth over R near $[P]$; see also [9, Lemma 2.5]. Since R is complete, hence henselian, its k-point $[P]$ lifts to an R-point. The universal subscheme gives $\mathscr{Z}$. $\square$

One may instead use Fogarty's relative smoothness theorem for Hilbert schemes of points on a smooth family of surfaces over a discrete valuation ring [6, Theorem 2.9]. The local complete intersection argument isolates precisely what is needed here. In particular, $\kappa(P)/k$ need not be separable.

Theorem 5.2 (The smooth case)**.** *Let $S \subset \mathbf{P}^3_k$ be a smooth cubic surface over an arbitrary field. If S has a zero-cycle of degree one, then $S(k) \neq \varnothing$.*

Proof. By Corollary 4.3, only positive characteristic remains. Choose a closed point P of degree d prime to three from the support of a zero-cycle of degree one. Apply Lemma 5.1, and put $K = \operatorname{Frac}(R)$. The finite scheme $\mathscr{Z}_K$ has degree d, so its fundamental cycle, together with the degree-three hyperplane intersection class, gives a zero-cycle of degree one on $\mathscr{S}_K$.

Corollary 4.3 gives a K-point of $\mathscr{S}_K$. Properness extends it to an R-point of $\mathscr{S}$, whose reduction is a k-point of S. $\square$

The argument retains the original residue field and includes characteristics two and three. Only the final rational point is specialized; the Pfaffian presentation and the isotropic incidence are used in characteristic zero.

Proposition 5.3. *Independently of Theorem 1.2, Voisin's characteristic-zero degree reduction implies that a smooth cubic surface over an arbitrary field with a zero-cycle of degree one has a closed point of degree one or four.*

Proof. In characteristic zero this is Voisin's theorem. In positive characteristic, lift a point of degree prime to three by Lemma 5.1 and apply Voisin's theorem to the smooth generic fiber. A rational point specializes by properness. Otherwise a degree-four point gives a K-point of the projective scheme $\operatorname{Hilb}^4(\mathscr{S}/R)$. Its extension over R specializes to a finite subscheme of degree four on S.

If its cycle has a point of degree one, we are done. A degree-two point also gives a rational point in arbitrary characteristic: its scheme-theoretic linear span is a k-line, and the cubic intersection either contains that line or leaves a residual cycle of degree one. This includes purely inseparable quadratic points. If neither degree occurs, the effective degree-four cycle is a single closed point of degree four. $\square$

Remark 5.4. The existence equivalence of Proposition 2.3 also holds for smooth cubic surfaces over arbitrary fields. In positive characteristic, if $S(k) = \varnothing$, quadratic secant descent excludes degree-two points, so an effective cycle of degree four or five is a single closed point. Lift it by Lemma 5.1 and apply Proposition 2.3 on the smooth generic fiber. The resulting effective cycle either yields a rational point, which specializes, or is a closed point of the other degree. Such a point specializes through the corresponding projective Hilbert scheme to a finite subscheme of that degree on S. If $S(k) \neq \varnothing$, multiples of a rational point give both degrees.

This transfers existence of effective cycles; it does not assert a positive-characteristic version of Voisin's Chow-group formulas.

6 Removing the smoothness hypothesis

Let $X = V(F) \subset \mathbf{P}^3_k$ for a nonzero homogeneous cubic F, with no assumption on k or on the singularities of X. If F is reducible over k, it has a linear factor ℓ, so X contains the k-plane $V(\ell)$ and $X(k) \neq \varnothing$. Write X_{sm} for the smooth locus over k. We first handle the possibility that a zero-cycle of degree one is supported entirely in the singular locus.

Lemma 6.1. *Let L/k be a finite extension with $3 \nmid [L : k]$. If $X(L) \neq \varnothing$, then either $X(k) \neq \varnothing$ or $X_{\mathrm{sm}}(L) \neq \varnothing$.*

Proof. Choose $p \in X(L)$. Its multiplicity on the hypersurface X_L is one, two, or three. Multiplicity one means that p is smooth.

Suppose the multiplicity is two. In coordinates over L with $p = [1 : 0 : 0 : 0]$, the equation is

$$F = x_0 Q(x_1, x_2, x_3) + C(x_1, x_2, x_3), \qquad Q \neq 0,$$

where Q and C have degrees two and three. There exists $a \in L^3$ with $Q(a) \neq 0$, even over a finite field: the values $Q(e_i)$ and $Q(e_i + e_j) - Q(e_i) - Q(e_j)$ recover all coefficients of Q. The condition $Q(a) \neq 0$ excludes directions whose lines through p are contained in X_L, in particular directions in any plane component through p. The point

$$q = \left[-\frac{C(a)}{Q(a)} : a_1 : a_2 : a_3 \right]$$

lies on X_L and satisfies $\partial F/\partial x_0(q) = Q(a) \neq 0$. It is therefore smooth. This is the usual projection from a double point; see [8, Section 5]. The coordinate argument applies unchanged to reducible and nonreduced hypersurfaces.

Suppose now that p has multiplicity three. A vertex of a cubic cone has multiplicity three and is therefore covered by this case. In coordinates with $p = [1 : 0 : 0 : 0]$, the equation is independent of x_0. In the original coordinates, a nonzero vector $v \in L^4$ representing p consequently satisfies

$$F(x + sv) = F(x) \quad \text{in } L[x_0, x_1, x_2, x_3, s]. \tag{6.1}$$

If $\operatorname{char} k \neq 3$, consider the k-linear map

$$D_F : k^4 \longrightarrow k[x_0, x_1, x_2, x_3]_2, \qquad w \longmapsto D_w F = \sum_{i=0}^{3} w_i \frac{\partial F}{\partial x_i}.$$

Its kernel becomes nonzero over L, hence contains a nonzero vector $w \in k^4$. Euler's identity gives $0 = (D_w F)(w) = 3F(w)$, so $[w] \in X(k)$.

If $\operatorname{char} k = 3$, the hypothesis on $[L : k]$ implies that L/k is separable. Normalize v so that one coordinate is one. Every conjugate $\sigma(v)$ satisfies (6.1), and directions satisfying this identity are closed under addition, since translations compose. Thus

$$w = \operatorname{Tr}_{L/k}(v) = \sum_{\sigma : L \hookrightarrow \overline{k}} \sigma(v) \in k^4$$

also satisfies (6.1). Its normalized coordinate is $[L : k] \neq 0$ in k, so $w \neq 0$. Setting $x = 0$ and $s = 1$ gives $F(w) = 0$. Finally, smoothness commutes with field extension, so the smooth points constructed on X_L belong to $X_{\mathrm{sm}}(L)$. ∎

Proposition 6.2 (Reduction to a smooth cubic over a Laurent-series field)**.** *Suppose every smooth cubic surface over $k((t))$ with a zero-cycle of degree one has a $k((t))$-rational point. Then every cubic surface over k with a zero-cycle of degree one has a k-rational point.*

Proof. Choose a closed point of degree d prime to three from the given zero-cycle, and let L/k be its residue extension. By Lemma 6.1, we may assume that there is a point $q \in X_{\mathrm{sm}}(L)$.

Choose a homogeneous cubic G defining a smooth surface over k. For example, one may take

$$G = \begin{cases} x_0^3 + x_1^3 + x_2^3 + x_3^3, & \operatorname{char} k \neq 3, \\ x_0^3 + x_0x_1^2 + x_1x_2^2 + x_2x_3^2, & \operatorname{char} k = 3. \end{cases}$$

In characteristic three, simultaneous vanishing of the partial derivatives forces $x_1 = x_2 = x_3 = 0$, and $[1:0:0:0]$ is not on $V(G)$. In the other characteristics the partial derivatives have no common projective zero.

Put $R = k[[t]]$, $K = k((t))$, and

$$\mathscr{X} = V\big((1-t)F + tG\big) \subset \mathbf{P}^3_R.$$

This is a flat projective family with special fiber X and smooth generic fiber $Y = \mathscr{X}_K$. Indeed, the image in $\mathbf{A}^1_k$ of the closed locus defined by the pencil and its partial derivatives is closed by properness and excludes $t = 1$, so it excludes the generic point.

The point q lifts to $\mathscr{X}(L[[t]])$ by Hensel's lemma: on an affine chart through q, hold all but a coordinate with nonzero partial derivative constant and lift the resulting simple root. Equivalently, apply the smooth-point lifting statement of [9, Lemma 2.4]. Its generic point belongs to $Y(L((t)))$. A k-basis of L is also a K-basis of $M = L((t))$, so $[M:K] = d$, without any separability assumption. If y denotes the image in Y, the pushforward cycle is

$$\zeta = [M : \kappa(y)]\,[y], \qquad \deg_K \zeta = d.$$

Combining ζ with the degree-three hyperplane intersection class gives a zero-cycle of degree one on Y. The assumed smooth theorem over K yields a point in $Y(K)$. Properness of $\mathscr{X}/R$ extends it to an R-point, whose reduction belongs to $X(k)$. □

Proof of Theorem 1.1. Apply Proposition 6.2, using Theorem 5.2 over $k((t))$. □

The reduction lifts only a smooth point over the chosen extension; it does not require a zero-cycle supported at singular points to lift to a smoothing. The final rational point may specialize to a singular point. In particular, the argument includes geometrically reducible and nonreduced cubic surfaces over finite and imperfect fields.

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